

Here are **perfect exam notes (ready to write)** for your topic **LEX & YACC** 📌

✍️ LEX and YACC (Exam Notes)

♦ LEX (Lexical Analyzer)

👉 Definition:

LEX is a tool used to **convert input stream into tokens**.

It identifies meaningful patterns (lexemes) using **regular expressions**.

♦ Functions of LEX

- Reads input **character by character**
 - Groups characters into **tokens**
 - Ignores spaces and comments
 - Sends tokens to parser (YACC)
-

♦ Structure of LEX Program

declarations

%%

translation rules

%%

auxiliary functions

♦ Example

%%

[0-9]+ { printf("NUMBER"); }

[a-zA-Z]+ { printf("IDENTIFIER"); }

%%

◆ Important Variables

- `yytext` → stores matched lexeme
 - `yylen` → length of lexeme
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◆ Working of LEX

1. Read input
 2. Match longest pattern
 3. Execute action
 4. Return token
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YACC (Yet Another Compiler Compiler)

◆ Definition

- 👉 YACC is a tool used to **analyze syntax (structure)** of tokens.
 - 👉 It generates a **parser**.
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◆ Functions of YACC

- Checks grammar rules
 - Detects syntax errors
 - Builds parse tree
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◆ Working of YACC

1. Takes grammar file (.y)
 2. Generates parser code
 3. Calls `yylex()` to get tokens
 4. Validates structure
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◆ Example

$S \rightarrow id = E$
 $E \rightarrow E + T \mid T$

Flow of LEX & YACC

Input $\rightarrow$ LEX $\rightarrow$ Tokens $\rightarrow$ YACC $\rightarrow$ Output

◆ Difference Between LEX and YACC

Feature	LEX	YACC
Work	Tokenization	Parsing
Input	Characters	Tokens
Output	Tokens	Syntax structure
Method	Regular expressions	Context-free grammar

👉 LEX performs lexical analysis by generating tokens, while YACC performs syntax analysis by checking grammar.

Memory Trick

👉 “LEX splits, YACC checks”

Exam Tip

👉 Write:

- Definition
- Structure
- Working
- Difference table

👉 This is enough for **10 marks** 100

If you want, I can give you **very short revision notes (1-page before exam)** 👉